

POWER BI ASSIGNMENT - 2 DOCUMENTATION

Module-2

Power BI Assignment 2 – DAX & Data Visualization

Project Title

E-Commerce Sales Data Analysis using Power BI

Submitted By

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ANB18 Batch

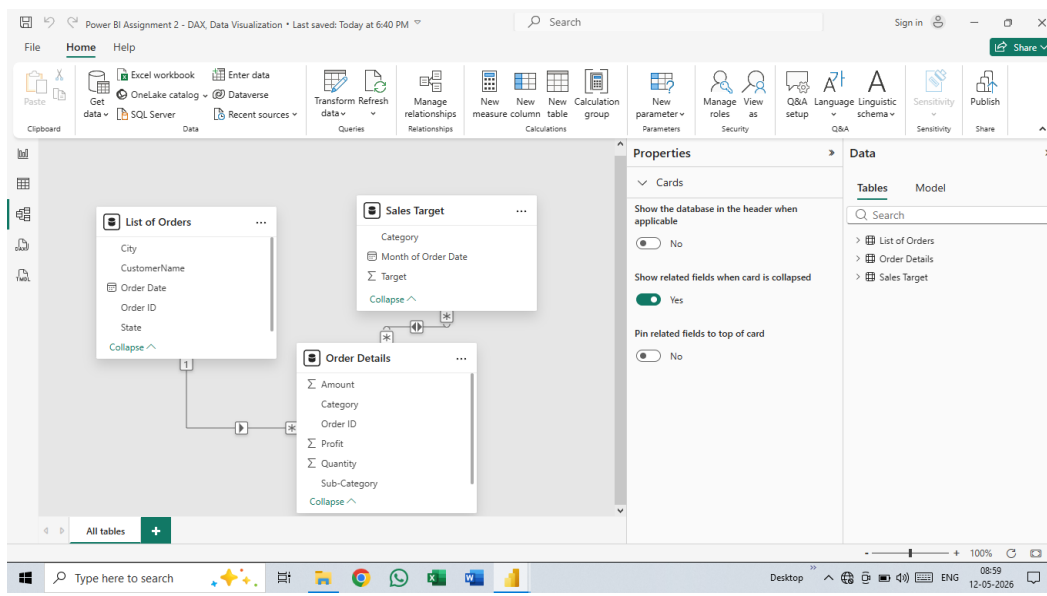
Dataset Overview :

- ❖ List of Order Table
- ❖ Order Details Table
- ❖ Sales Target Table

Transformation Steps :

- To Enhance my Data Analysis, I First loaded my CSV File Dataset into Power BI using the Load.
- After that, I used DAX formulas to make the Necessary Changes to my Data.
- Subsequently, to Establish Relationships between my tables, I defined the necessary Relationship within the Data Modeling View.
- Through this, I can utilize my data to create Interactive Visuals-such as Donut Chart, Line Chart, and Column Chart-as well as KPI, Slicer, and Filter.

Data Model



- ❖ To Significantly Simplify and accelerate my data Analysis-and to retrieve the specific data I require in various ways-I utilized data model relationships to integrate my 'List of Orders', 'Order Details', and 'Sales Target' tables,

Establishing ONE to MANY and MANY to MANY Relationships between them.

Calculated Columns :

❖ **DAX Formula :** Revenue = 'Order Details'[Amount] * 'Order Details'[Quantity]

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File Home Help Table tools Column tools

Name: Revenue Format: Whole number Summarization: Sum Data category: Uncategorized

Structure: Columns: Order ID, Amount, Profit, Quantity, Category, Sub-Category, Category Type, Revenue, Sales Category

Table: Order Details (1,500 rows) Column: Revenue (845 distinct values)

Order ID	Amount	Profit	Quantity	Category	Sub-Category	Category Type	Revenue	Sales Category
B-25602	561	212	3	Clothing	Saree	Clothing-Saree	1683	Above Average
B-25602	119	-5	8	Clothing	Saree	Clothing-Saree	952	Below Average
B-25603	193	-166	3	Clothing	Saree	Clothing-Saree	579	Below Average
B-25604	157	5	9	Clothing	Saree	Clothing-Saree	1413	Below Average
B-25605	75	0	7	Clothing	Saree	Clothing-Saree	525	Below Average
B-25609	25	-5	4	Clothing	Saree	Clothing-Saree	100	Below Average
B-25610	43	0	3	Clothing	Saree	Clothing-Saree	129	Below Average
B-25611	160	-59	2	Clothing	Saree	Clothing-Saree	320	Below Average
B-25613	1603	0	9	Clothing	Saree	Clothing-Saree	14427	Above Average
B-25619	353	90	8	Clothing	Saree	Clothing-Saree	2824	Above Average
B-25622	534	0	3	Clothing	Saree	Clothing-Saree	1602	Above Average
B-25623	149	-87	4	Clothing	Saree	Clothing-Saree	596	Below Average
B-25625	635	-349	5	Clothing	Saree	Clothing-Saree	3175	Above Average
B-25628	24	-9	4	Clothing	Saree	Clothing-Saree	96	Below Average
B-25633	711	-8	4	Clothing	Saree	Clothing-Saree	2844	Above Average
B-25635	382	30	3	Clothing	Saree	Clothing-Saree	1146	Above Average
B-25636	637	113	5	Clothing	Saree	Clothing-Saree	3185	Above Average
B-25640	122	-47	4	Clothing	Saree	Clothing-Saree	488	Below Average
B-25646	20	-8	2	Clothing	Saree	Clothing-Saree	40	Below Average
B-25647	42	-6	4	Clothing	Saree	Clothing-Saree	168	Below Average
B-25648	55	-26	4	Clothing	Saree	Clothing-Saree	220	Below Average
B-25648	130	-41	4	Clothing	Saree	Clothing-Saree	520	Below Average
B-25650	211	-105	2	Clothing	Saree	Clothing-Saree	422	Below Average

❖ The Calculated Column Proved to be Extremely Useful; by Creating it and utilizing the DAX formula, I Was Able to generate a new 'Revenue' Column by multiplying the 'Order Detail' [Amount] and 'Order Detail' [Quantity] Fields to Calculate the Total Revenue.

❖ **DAX Formula :** Sales Category = IF('Order Details'[Amount]>AVERAGE('Order Details'[Amount]), "Above Average", "Below Average")

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File Home Help Table tools Column tools

Name: Sales Category Format: Text Summarization: Don't summarize Data category: Uncategorized

Structure: Columns: Order ID, Amount, Profit, Quantity, Category, Sub-Category, Category Type, Revenue, Sales Category

Table: Order Details (1,500 rows) Column: Sales Category (2 distinct values)

Order ID	Amount	Profit	Quantity	Category	Sub-Category	Category Type	Revenue	Sales Category
B-25602	561	212	3	Clothing	Saree	Clothing-Saree	1683	Above Average
B-25602	119	-5	8	Clothing	Saree	Clothing-Saree	952	Below Average
B-25603	193	-166	3	Clothing	Saree	Clothing-Saree	579	Below Average
B-25604	157	5	9	Clothing	Saree	Clothing-Saree	1413	Below Average
B-25605	75	0	7	Clothing	Saree	Clothing-Saree	525	Below Average
B-25609	25	-5	4	Clothing	Saree	Clothing-Saree	100	Below Average
B-25610	43	0	3	Clothing	Saree	Clothing-Saree	129	Below Average
B-25611	160	-59	2	Clothing	Saree	Clothing-Saree	320	Below Average
B-25613	1603	0	9	Clothing	Saree	Clothing-Saree	14427	Above Average
B-25619	353	90	8	Clothing	Saree	Clothing-Saree	2824	Above Average
B-25622	534	0	3	Clothing	Saree	Clothing-Saree	1602	Above Average
B-25623	149	-87	4	Clothing	Saree	Clothing-Saree	596	Below Average
B-25625	635	-349	5	Clothing	Saree	Clothing-Saree	3175	Above Average
B-25628	24	-9	4	Clothing	Saree	Clothing-Saree	96	Below Average
B-25633	711	-8	4	Clothing	Saree	Clothing-Saree	2844	Above Average
B-25635	382	30	3	Clothing	Saree	Clothing-Saree	1146	Above Average
B-25636	637	113	5	Clothing	Saree	Clothing-Saree	3185	Above Average
B-25640	122	-47	4	Clothing	Saree	Clothing-Saree	488	Below Average
B-25646	20	-8	2	Clothing	Saree	Clothing-Saree	40	Below Average
B-25647	42	-6	4	Clothing	Saree	Clothing-Saree	168	Below Average
B-25648	55	-26	4	Clothing	Saree	Clothing-Saree	220	Below Average
B-25648	130	-41	4	Clothing	Saree	Clothing-Saree	520	Below Average
B-25650	211	-105	2	Clothing	Saree	Clothing-Saree	422	Below Average

- ❖ To Create a Calculated Column Named “Sales Category”, I utilized a DAX Formula to analyze the “Amount” Column; Specifically, I created a Separate column that labels values as “Above Average” if they exceed the Average Amount, and “Below Average” if they fall Below it.
- ❖ **DAX Formula :** Category Type = ‘Order Details’[Category] & “-” & ‘Order Details’ [Sub-Category]

Order ID	Amount	Profit	Quantity	Category	Sub-Category	Category Type	Revenue	Sales Category
B-25602	561	212	3	Clothing	Saree	Clothing-Saree	1683	Above Average
B-25602	119	-5	8	Clothing	Saree	Clothing-Saree	952	Below Average
B-25603	183	-166	3	Clothing	Saree	Clothing-Saree	579	Below Average
B-25604	157	5	9	Clothing	Saree	Clothing-Saree	1413	Below Average
B-25605	75	0	7	Clothing	Saree	Clothing-Saree	523	Below Average
B-25609	25	-5	4	Clothing	Saree	Clothing-Saree	100	Below Average
B-25610	43	0	3	Clothing	Saree	Clothing-Saree	129	Below Average
B-25611	160	-59	2	Clothing	Saree	Clothing-Saree	320	Below Average
B-25613	1803	0	9	Clothing	Saree	Clothing-Saree	14427	Above Average
B-25619	313	90	8	Clothing	Saree	Clothing-Saree	2624	Above Average
B-25621	534	0	3	Clothing	Saree	Clothing-Saree	1652	Above Average
B-25623	149	-87	4	Clothing	Saree	Clothing-Saree	596	Below Average
B-25625	835	-349	3	Clothing	Saree	Clothing-Saree	2173	Above Average
B-25626	24	-9	4	Clothing	Saree	Clothing-Saree	96	Below Average
B-25631	711	-8	4	Clothing	Saree	Clothing-Saree	2644	Above Average
B-25635	382	30	3	Clothing	Saree	Clothing-Saree	1146	Above Average
B-25636	837	113	5	Clothing	Saree	Clothing-Saree	2185	Above Average
B-25640	122	-47	4	Clothing	Saree	Clothing-Saree	498	Below Average
B-25646	20	-8	2	Clothing	Saree	Clothing-Saree	40	Below Average
B-25647	42	-8	4	Clothing	Saree	Clothing-Saree	168	Below Average
B-25648	55	-26	4	Clothing	Saree	Clothing-Saree	200	Below Average
B-25648	130	-41	4	Clothing	Saree	Clothing-Saree	520	Below Average
B-25656	711	-195	3	Clothing	Saree	Clothing-Saree	477	Below Average

- ❖ To Create a Calculated Column Named “Category Type”, I utilized DAX formula to Concatenate the Category and Sub-Category fields; Specifically, I used the ‘&’ operator to join the Category, a (‘-’), and the Sub-Category together.

Calculated Measures :



1. Order Count
2. Average Profit in Delhi
3. Year-to-Date (YTD)

1. Order Count

❖ **DAX Formula :** Total Order = Count ('Order Details'[Order ID])

The screenshot shows the Power BI Desktop interface. The 'Table tools' ribbon is active, and the 'Measure tools' section is visible. The 'Name' field shows the formula: **Total Orders = COUNT('Order Details'[Order ID])**. The 'Data' pane on the right shows the 'List of Orders' table selected. The main table view displays a list of orders with columns: Order ID, Amount, Profit, Quantity, Category, Sub-Category, Category Type, Revenue, and Sales Category. The table contains 1,500 rows of data.

Order ID	Amount	Profit	Quantity	Category	Sub-Category	Category Type	Revenue	Sales Category
B-25602	561	212	3	Clothing	Saree	Clothing-Saree	1683	Above Average
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B-25605	75	0	7	Clothing	Saree	Clothing-Saree	525	Below Average
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B-25610	43	0	3	Clothing	Saree	Clothing-Saree	129	Below Average
B-25611	160	-59	2	Clothing	Saree	Clothing-Saree	320	Below Average
B-25613	1603	0	9	Clothing	Saree	Clothing-Saree	14427	Above Average
B-25619	353	90	8	Clothing	Saree	Clothing-Saree	2824	Above Average
B-25622	534	0	3	Clothing	Saree	Clothing-Saree	1602	Above Average
B-25623	149	-87	4	Clothing	Saree	Clothing-Saree	596	Below Average
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B-25628	24	-9	4	Clothing	Saree	Clothing-Saree	96	Below Average
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B-25640	122	-47	4	Clothing	Saree	Clothing-Saree	488	Below Average
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B-25648	130	-41	4	Clothing	Saree	Clothing-Saree	520	Below Average
B-25650	211	-105	2	Clothing	Saree	Clothing-Saree	422	Below Average
B-25650	31	-2	2	Clothing	Saree	Clothing-Saree	62	Below Average
B-25650	512	-225	5	Clothing	Saree	Clothing-Saree	2560	Above Average

2. Average Profit in Delhi

❖ **DAX Formula :** Average Profit in Delhi = Calculated (Average ('Order Details'[Profit]), 'List of Orders' [city]="Delhi")

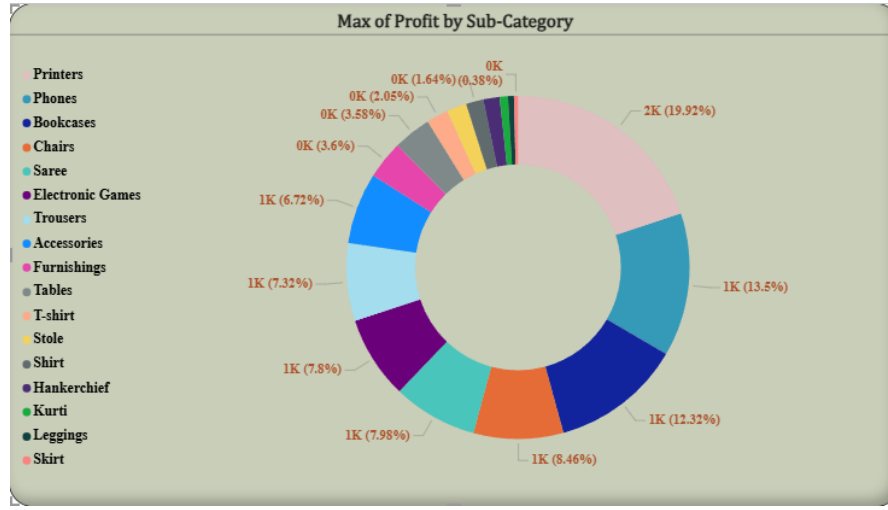
The screenshot shows the Power BI Desktop interface. The 'Table tools' ribbon is active, and the 'Measure tools' section is visible. The 'Name' field shows the formula: **Average Profit in Delhi = CALCULATE(AVERAGE('Order Details'[Profit]), 'List of Orders'[City]="Delhi")**. The 'Data' pane on the right shows the 'List of Orders' table selected. The main table view displays a list of orders with columns: Order ID, Amount, Profit, Quantity, Category, Sub-Category, Category Type, Revenue, and Sales Category. The table contains 1,500 rows of data.

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B-25648	130	-41	4	Clothing	Saree	Clothing-Saree	520	Below Average
B-25650	211	-105	2	Clothing	Saree	Clothing-Saree	422	Below Average
B-25650	31	-2	2	Clothing	Saree	Clothing-Saree	62	Below Average
B-25650	512	-225	5	Clothing	Saree	Clothing-Saree	2560	Above Average

3. Year-to-Date (YTD) Sales

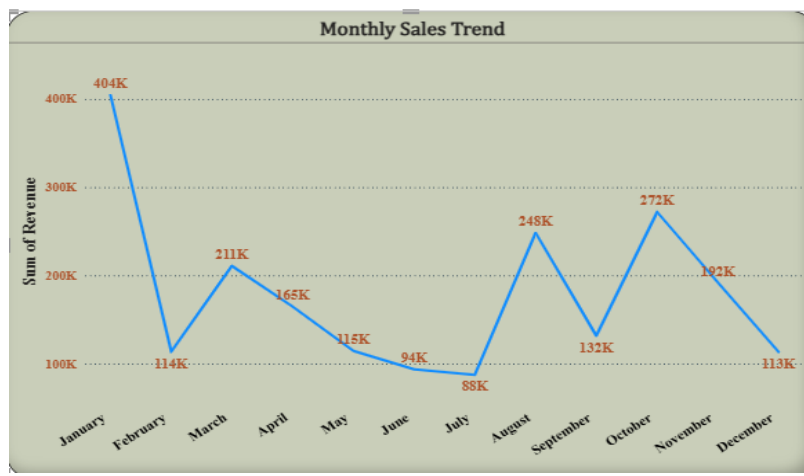
❖ **DAX Formula :** YTD = TOTALYTD(SUM('Order Details' [Revenue]), 'List of Orders' [Order Date])

Max of Profit by Sub-Category



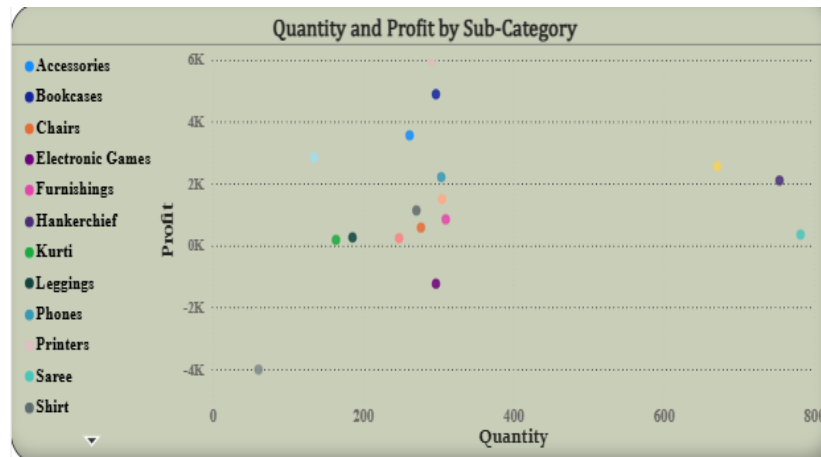
- ❖ By utilizing a Donut Chart, we can Visualize the Maximum Profit Generated by each sub-category. This Chart Clearly reveals that “**Printers**” have yielded the highest profit. When compared to the other sub-categories, printers stand at the top position, accounting for 19.92% of the total earnings.

Monthly Sales Trend



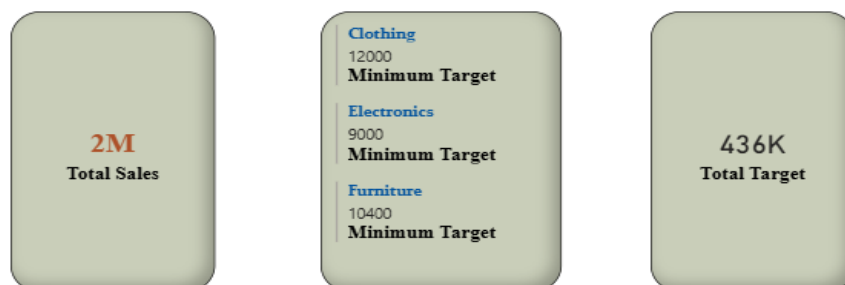
- ❖ By utilizing a line Chart to analyze monthly Sales trends, we can gain insights into the overall performance of our Business. Specifically, the line Chart Enables us to identify which months generated the highest sales volume and which months experienced lower sales. Upon Viewing the visualization, it becomes immediately apparent that the month of January Recorded the highest Saales Activity.

Quantity And Profit by Sub-Category



- ❖ By utilizing a scatter Chart, we can visualize which specific product has generated the highest profit and the corresponding quantity sold. Specifically, by assigning the 'Sub-Category' to the Chart's Legend, we can Clearly identify which products have generated the most revenue and the volume of sales achieved. In this context-and when compared to other sub-categories 'Bookcases' 297units emerged as the top selling item, which also generating the highest profit ₹4888.

Total Sales Amount and Target



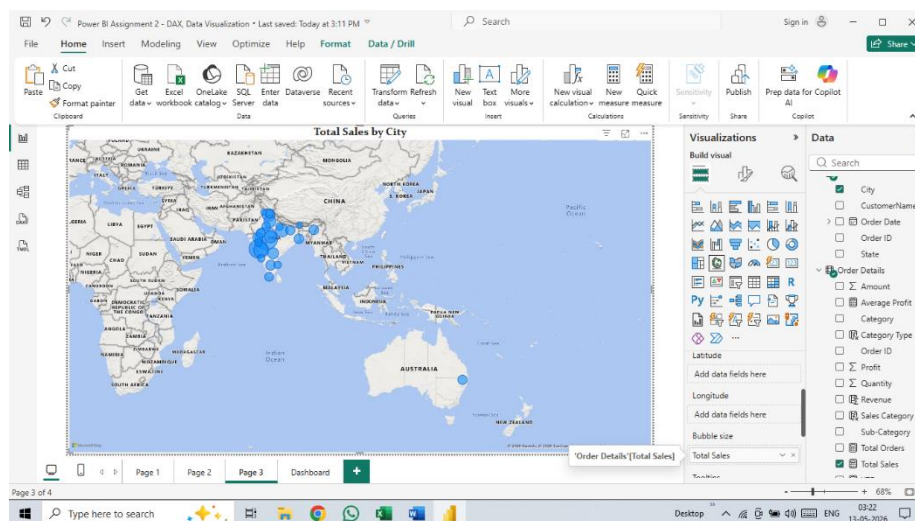
- ❖ By utilizing the Card visual, we can observe that the total Sales amount is 2Million and the total Target is 436K. Simultaneously, by employing the Multi-Row Card visual, we can view the minimum targets for the Cloth, Electronics, and Furniture categories; through this, it is evident that the minimum target for the Cloth Category is higher than that of the other Categories.

Sales Performance Matrix

Category	Clothing		Electronics		Furniture		Total	
Month	Total Sales	Total Target	Total Sales	Total Target	Total Sales	Total Target	Total Sales	Total Target
January	86086	174000	179726	129000	138435	132900	404247	435900
February	32894	174000	49433	129000	31260	132900	113587	435900
March	78281	174000	83246	129000	49403	132900	210930	435900
April	63491	174000	34544	129000	66764	132900	164799	435900
May	39479	174000	51941	129000	23115	132900	114535	435900
June	38023	174000	36328	129000	19394	132900	93745	435900
July	19609	174000	38142	129000	29884	132900	87635	435900
August	76377	174000	86094	129000	85690	132900	248161	435900
September	55240	174000	39669	129000	36924	132900	131833	435900
October	67726	174000	94525	129000	109785	132900	272036	435900
November	72885	174000	65207	129000	53959	132900	192051	435900
Total	664522	174000	816583	129000	665765	132900	2146870	435900

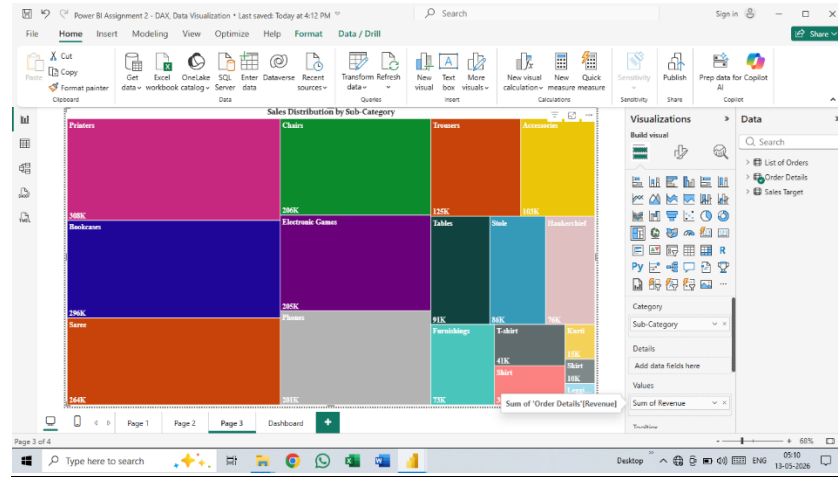
- ❖ By utilizing metrics to assess sales Performance, one can track-on a Monthly basis-the total sales for each category and determine whether the respective sales targets have been met. In this regard, the electronics Category Successfully achieved its target; sales for January were robust. Similarly, the furniture category also successfully met its monthly target for January.

Total Sales by City



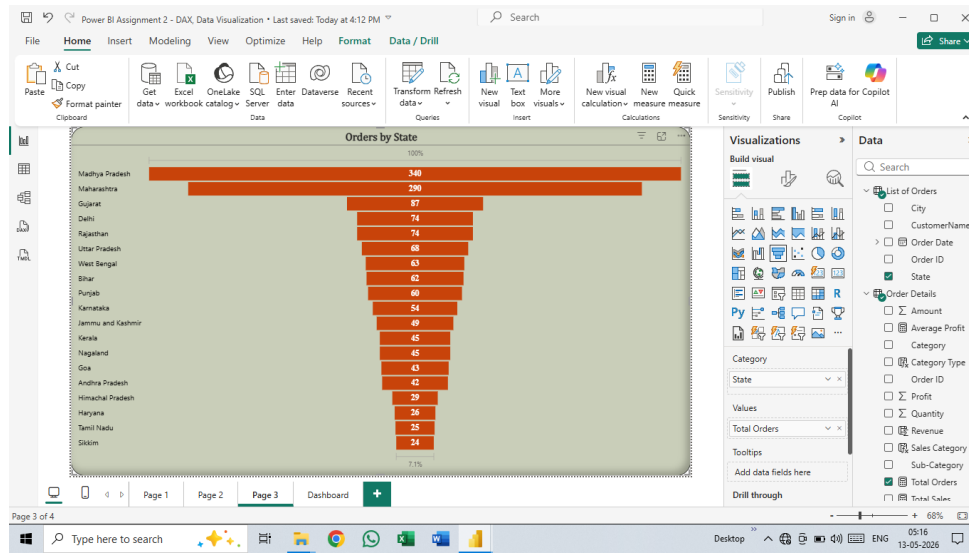
- ❖ By utilizing a geographic map, we are able to visualize the city-wise sales Performance Through this Map.

Sales Distribution by Sub-Category



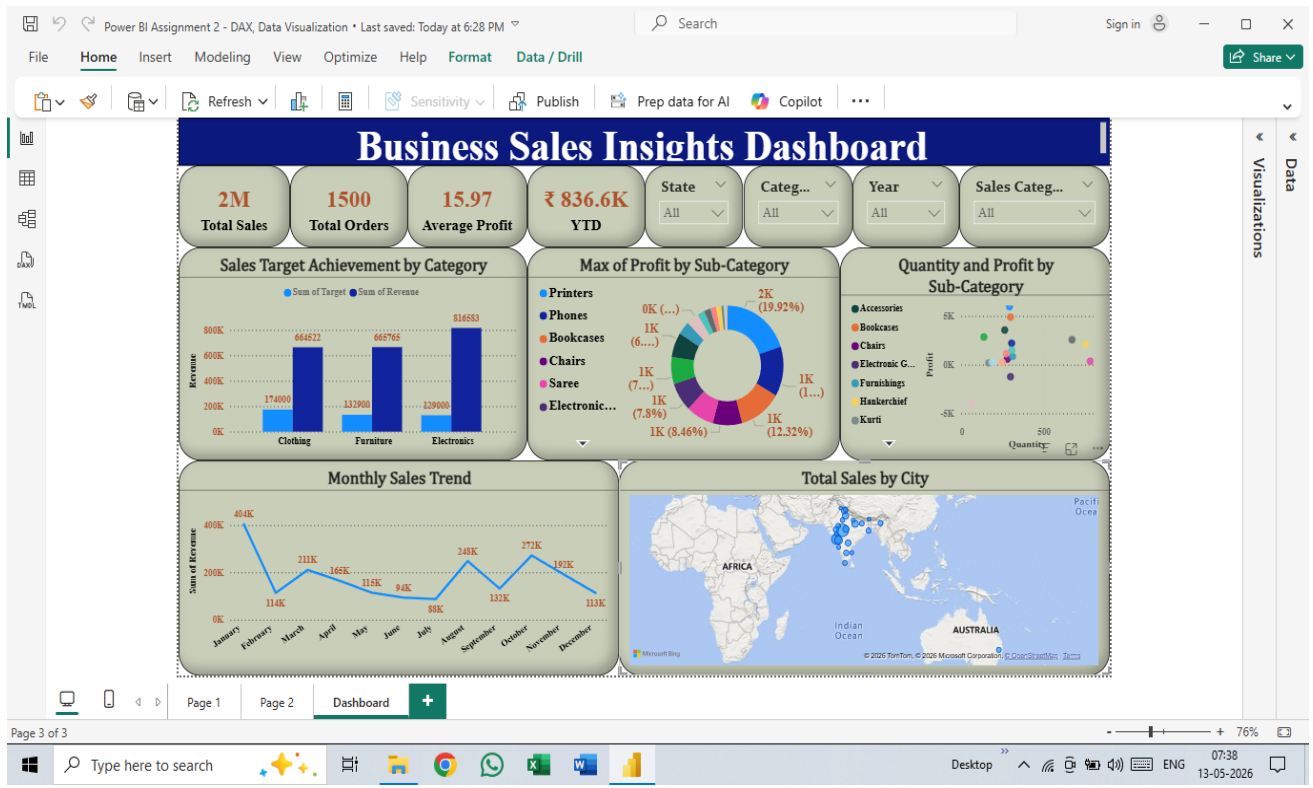
- ❖ A tree Map Allows us to identify which Sub-Categories have generated the Highest Sales volume. Based on this Visualization, we can observe that printers having Recorded the highest sales have consequently generated the greatest Revenue.

Orders By State



- ❖ Through this Funnel chart, we can observe which state generated the highest volume of orders; specifically, Madhya Pradesh recorded the Highest Sales.

DASHBOARD



- ✓ The Dashboard provides Clear Insights into business Sales Performance, profit distribution, and regional sales trends.
- ✓ Electronics and sub-categories like Phones and Printers contributed Significantly to overall revenue and Profit.
- ✓ The Analysis also Highlights the importance of Monitoring monthly trends and city-wise sales to improve decision-making and optimize future sales strategies.

Thank You